

MACHINE LEARNING

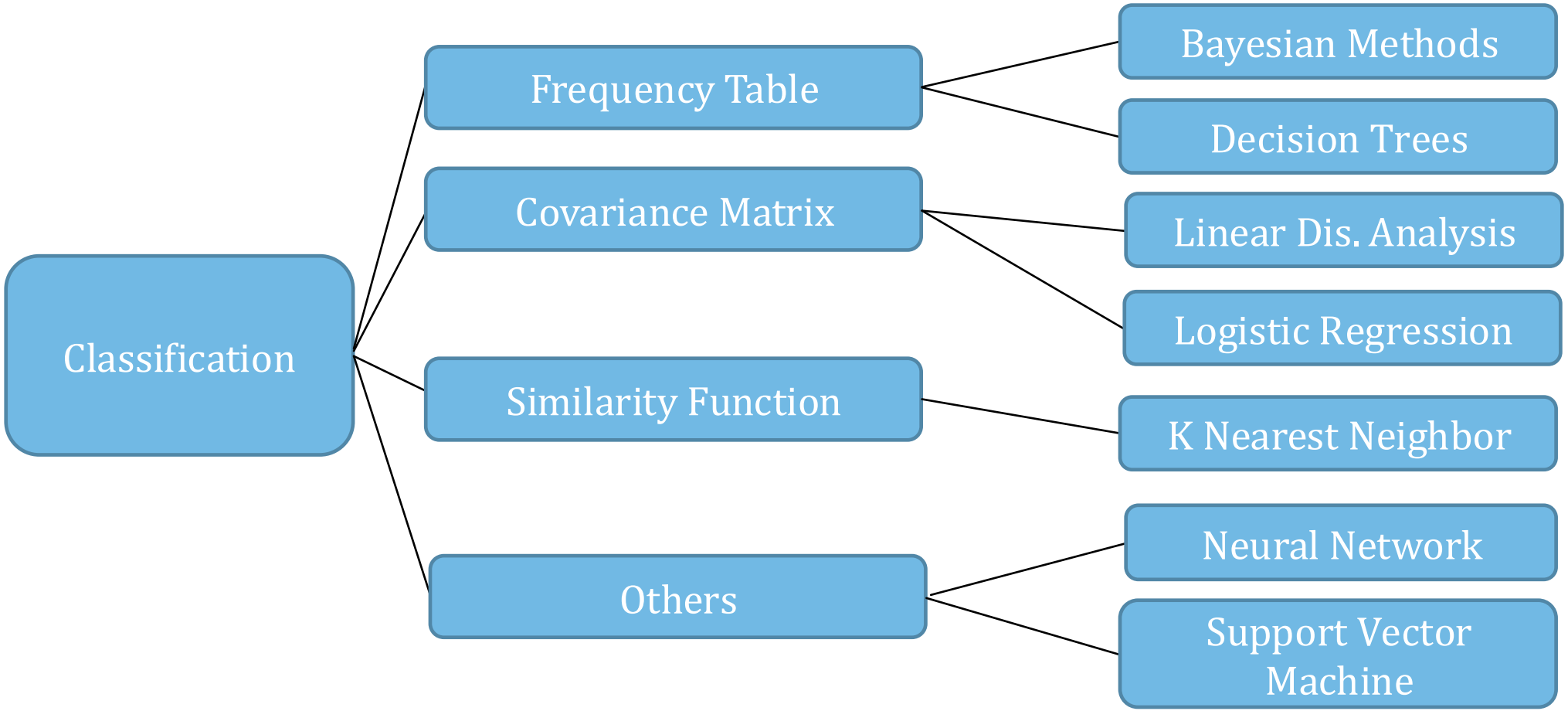
K-NEAREST NEIGHBOR, DECISION TREES & BOOSTING ALGORITHMS

NGO QUOC VIET-2025

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1. K-Nearst Neighbor.
2. Decision Trees.
3. Boosting algorithms.

INTRODUCTION



K-NEAREST-NEIGHBOR ALGORITHM

- K nearest neighbor (KNN) is a simple **unsupervised and non-parametric** algorithm that stores all available cases and classifies new cases based on a similarity measure (distance function)

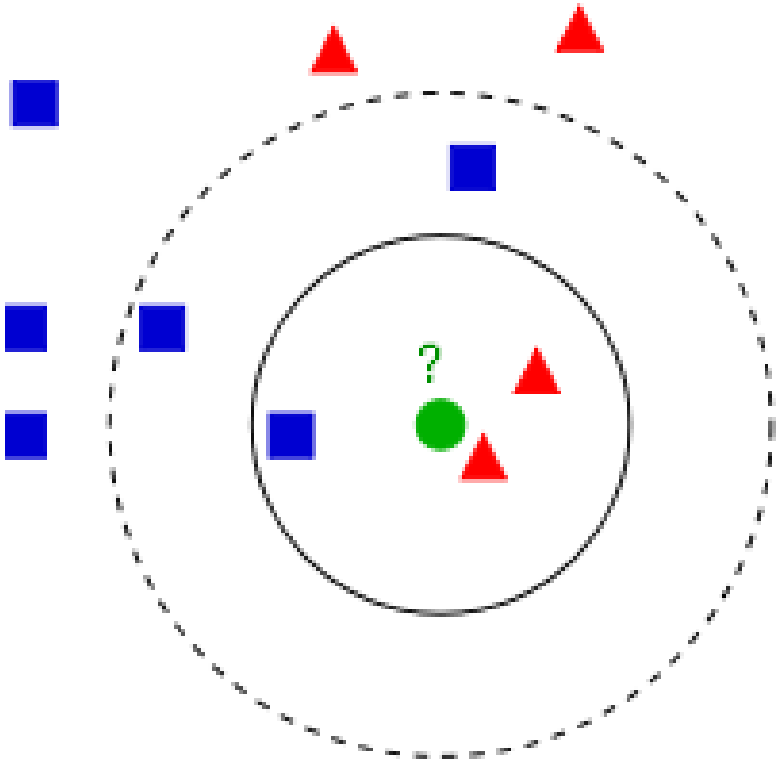
- KNN has been used in statistical estimation and pattern recognition since 1970's.

TRAINING KNN

No training is needed or training do nothing. Sometimes, training is normalizing data.

TESTING

Depends on value of k. If *k = 3* (solid line circle) it is assigned to the red triangles because there are 2 triangles and only 1 square inside the inner circle. If *k = 5* (dashed line circle) it is assigned to the blue squares (3 squares vs. 2 triangles inside the outer circle).



https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

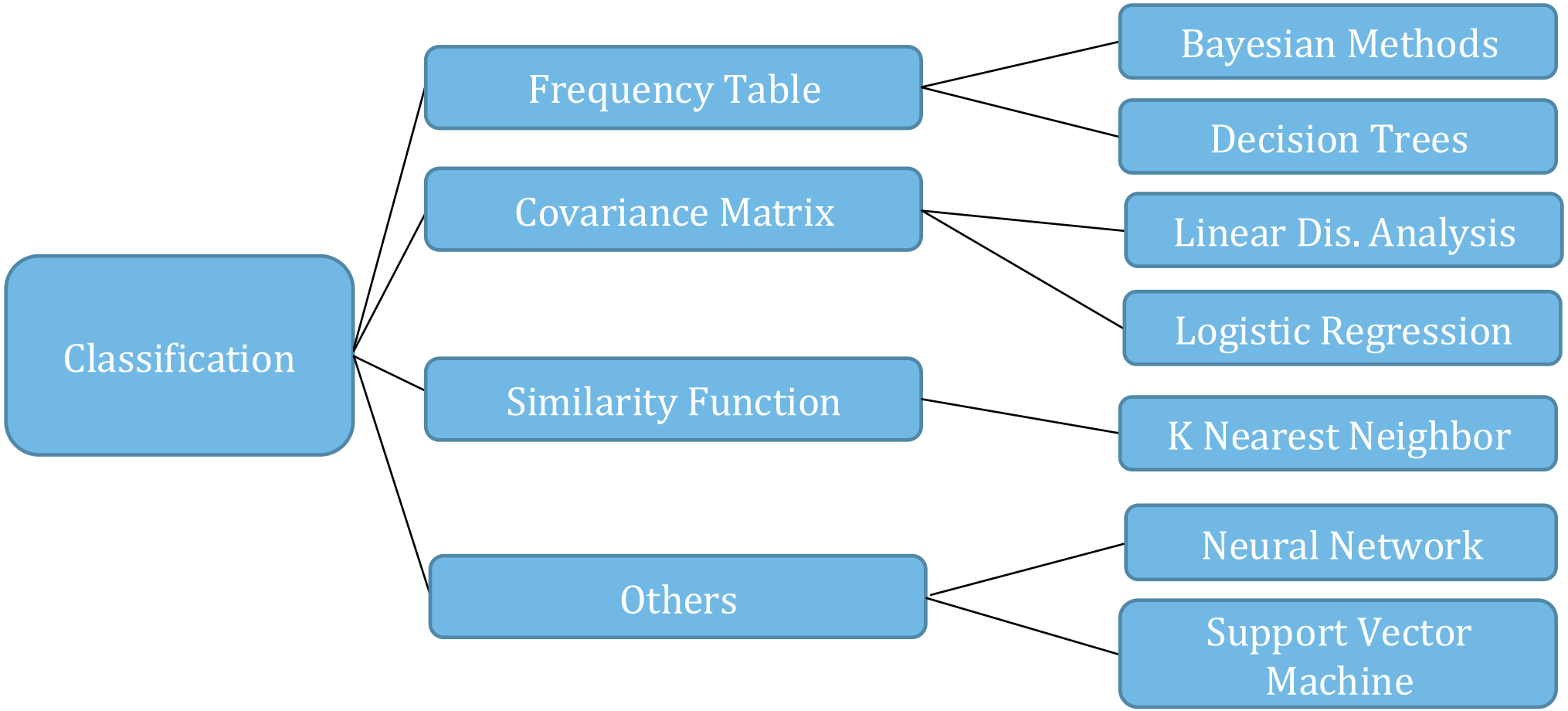
K-NEAREST-NEIGHBOR, MORE

- Unlike all the previous learning methods, kNN does not build model from the training data.
- To classify a test instance d, define k-neighborhood P as k nearest neighbors of d.
- Count number n of training instances in P that belong to class c_j
- Estimate $\Pr(c_j|d)$ as n/k.
- **No training is needed or training do nothing. Sometimes, training is normalizing data.**
- **Classification time is linear in training set size for each test case.**

K-NEAREST-NEIGHBOR SIMPLE CODE

https://www.datacamp.com/community/tutorials/k-nearest-neighbor-classification-scikit-learn

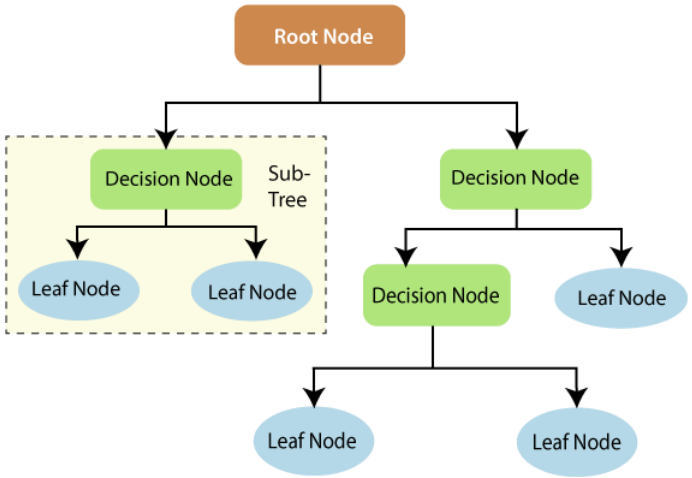
INTRODUCTION



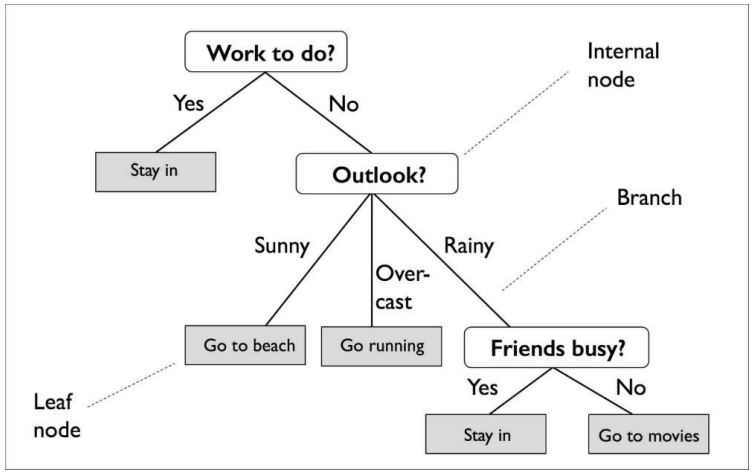
DECISION TREES

- Decision Tree algorithm belongs to the family of **supervised** and **non-parametric** learning algorithms.
- Can be used for **regression** and **classification** problems.
- There are two types:
  1. **Categorical Variable Decision Tree:** Decision Tree which has a categorical target variable then it called a **Categorical variable decision tree.**
  2. **Continuous Variable Decision Tree:** Decision Tree has a continuous target variable then it is called **Continuous Variable Decision Tree.**
- The goal of using a Decision Tree is to create a training model that can use to predict the class or value of the target variable by **learning simple decision rules** inferred from training data.

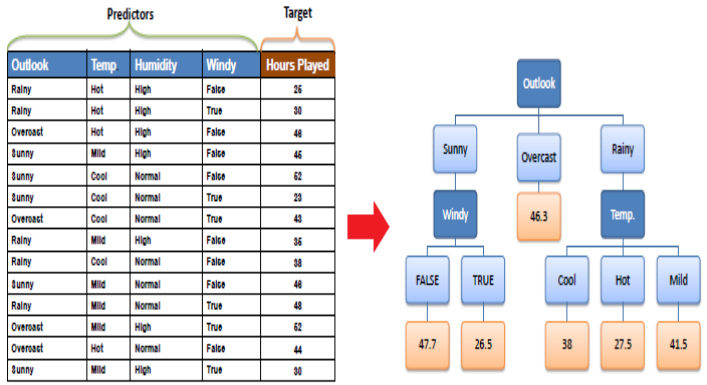
DECISION TREES



Classification Tree



Regression Tree



HOW DO DECISION TREES WORK

- Builds classification or regression models in the form of a tree structure.
- It breaks down a data set into smaller and smaller subsets. Splits are formed on a particular variable.
- The decision of making strategic splits heavily affects a tree's accuracy. The decision criteria are different for classification and regression trees.
- The final result is a tree with decision nodes and leaf nodes.
- The algorithm selection is also based on the type of target variables: ID3, C4.5, CART (Classification And Regression Tree), CHAID, MARS.

HOW DO DECISION TREES WORK

1. Select the best attribute using Attribute Selection Measures (ASM) to split the records.
   - Decision trees split on the feature and corresponding split point that results in the **largest information gain** for a given criterion (gini or entropy).
2. Make that attribute a decision node and breaks the dataset into smaller subsets.
3. Starts tree building by repeating this process recursively for each child until one of the condition will match:
   - All the tuples belong to the same attribute value.
   - There are no more remaining attributes.
   - There are no more instances.

ATTRIBUTE SELECTION MEASURES

- **Entropy**: measure of the randomness in the information. The higher the entropy, the harder it is to draw any conclusions from that information.

$$E(S) = \sum_{i=1}^{c} -p_i \log_2 p_i$$

- For multi attributes

$$E(T, X) = \sum_{c \in X} P(c)E(c)$$

		Play		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
				14

$$E(Play) = E(5/14, 9/14) = 0.94$$

$$E(Play, Outlook)$$
$$= P(sunny)E(3,2) + P(overcast)E(4,0)$$
$$+ P(rainy)E(2,3)$$

$$= (5/14) * 0.971 + \left(\frac{4}{14}\right) * 0.0 + \left(\frac{5}{14}\right) * 0.971 = 0.693$$

ATTRIBUTE SELECTION MEASURES

- **Information Gain**: measures how well a given attribute separates the training examples according to their target classification. ***Higher is better attribute to split***.

$$IG(T, X) = E(T) - E(T, X)$$

- **Gini Index**: measure of misclassification and is used when the data contain multi class labels. Gini is similar to entropy but it calculates much quicker than entropy. Algorithms like CART (Classification and Regression Tree) use Gini as an **impurity** parameter. ***Smaller is better***.

$$GI = 1 - \sum_{i=1}^{c} (p_i)^2$$

		Play		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
				14

$$IG(Play, Outlook) = 0.94 - 0.693 = 0.247$$

$$GI(Outlook == Sunny) = 1 - \left(\left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2\right)$$

ATTRIBUTE SELECTION MEASURES

- **Gain ratio**: overcomes the problem with information gain by taking into account the number of branches that would result before making the split.

$$GR = \frac{IG}{SplitInfo} = \frac{E(before) - \sum_{j=1}^{K} E(j, after)}{-\sum_{j=1}^{K} w_j \log_2 w_j}$$

ID code	Outlook	Temp.	Humidity	Windy	Play
A	Sunny	Hot	High	False	No
B	Sunny	Hot	High	True	No
C	Overcast	Hot	High	False	Yes
D	Rainy	Mild	High	False	Yes
E	Rainy	Cool	Normal	False	Yes
F	Rainy	Cool	Normal	True	No
G	Overcast	Cool	Normal	True	Yes
H	Sunny	Mild	High	False	No
I	Sunny	Cool	Normal	False	Yes
J	Rainy	Mild	Normal	False	Yes
K	Sunny	Mild	Normal	True	Yes
L	Overcast	Mild	High	True	Yes
M	Overcast	Hot	Normal	False	Yes
N	Rainy	Mild	High	True	No

	Info	IG	Split Info	GR
Outlook	0.693	0.940-0.693=0.247	Info([5, 4, 5])=1.577	0.247/1.577=**0.157**
Temp	0.911	0.940-0.911=0.029	Info([5, 6, 4])=1.565	0.029/1.565=0.019
Humidity	0.778	0.940-0.778=0.152	Info([7, 7])=1.000	0.152/1.000=0.152
Windy	0.892	0.940-0.892=0.048	Info([8, 6])=0.985	0.048/0.985=0.049

CLASSIFICATION WITH DECISION TREES

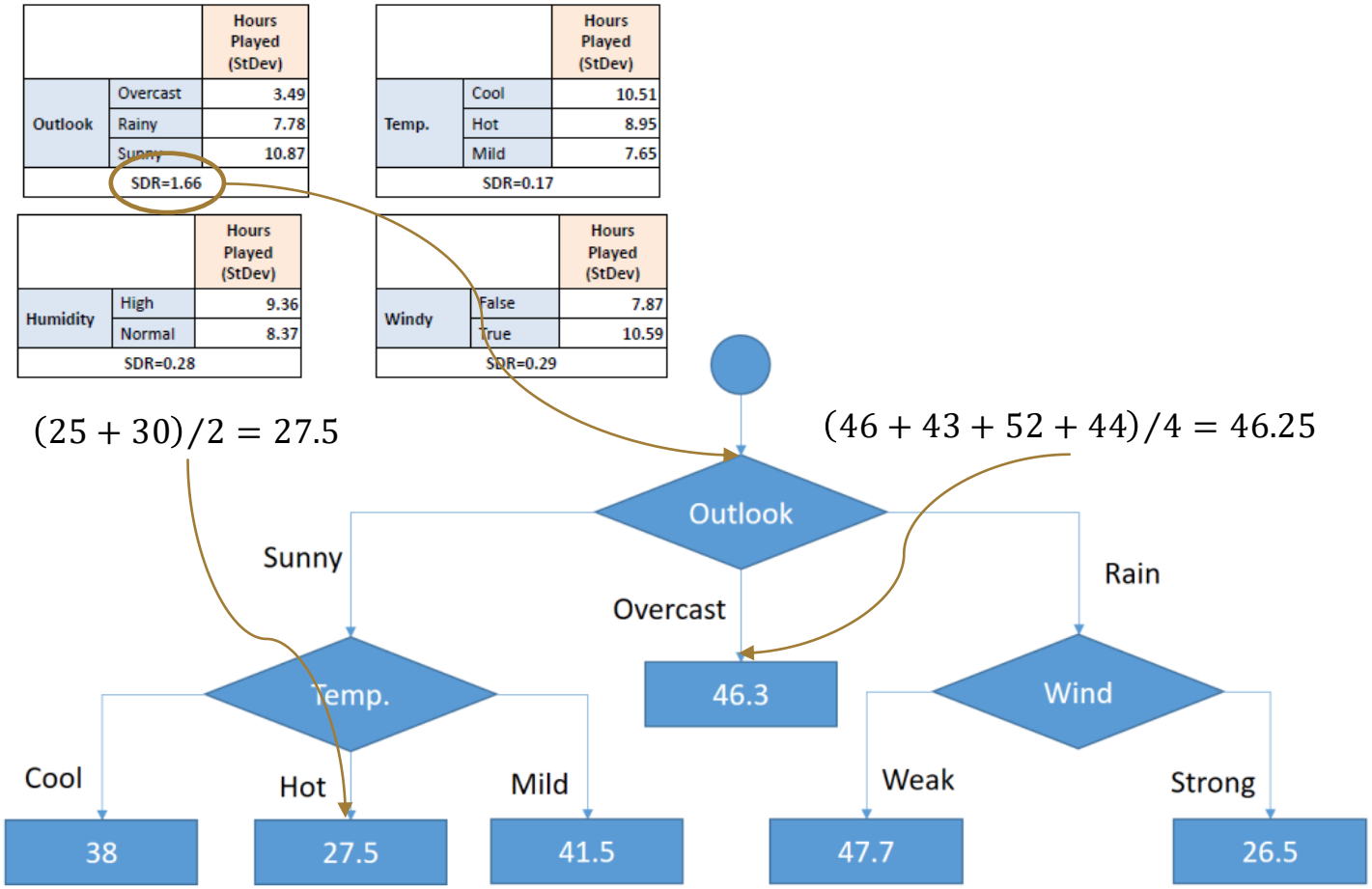
- Given testing data, move from the root to leaf node to get the prediction.
- https://www.youtube.com/watch?v=7VeUPuFGJHk.

REGRESSION WITH DECISION TREES

- Whenever, the target variable is numeric, we use Regression Tree instead of Classification Tree.
- Using GINI (Impurity measure of node, *smaller is better*), **S**tandard **D**eviation **R**eduction, **R**eduction of **V**ariance (*larger is better*), or **C**oefficient of **V**ariance to determine split node.
- The polular algorithms: CART.
- *Value of leaf node* in decision tree is mean of objects in node.

REGRESSION WITH DECISION TREES

Day	Outlook	Temp.	Humi.	Wind	Hours Playeed
1	Sunny	Hot	High	Weak	25
2	Sunny	Hot	High	Strong	30
3	Overcast	Hot	High	Weak	46
4	Rain	Mild	High	Weak	45
5	Rain	Cool	Normal	Weak	52
6	Rain	Cool	Normal	Strong	23
7	Overcast	Cool	Normal	Strong	43
8	Sunny	Mild	High	Weak	35
9	Sunny	Cool	Normal	Weak	38
10	Rain	Mild	Normal	Weak	46
11	Sunny	Mild	Normal	Strong	48
12	Overcast	Mild	High	Strong	52
13	Overcast	Hot	Normal	Weak	44
14	Rain	Mild	High	Strong	30



DECISION TREES EXPLAINED AND SIMPLE CODE

- https://www.youtube.com/watch?v=_L39rN6gz7Y
- https://www.datacamp.com/community/tutorials/decision-tree-classification-python

ADVANTAGES AND DISADVANTAGES OF DECISION TREES

Advantages

- It is easier to use.
- Fastest: finding the most significant variables, and their relationship .
- It requires very little effort to prepare the data.
- It does not get influenced by outliers and missing values.
- It is not concerned with the data type (categorical as well as numerical).

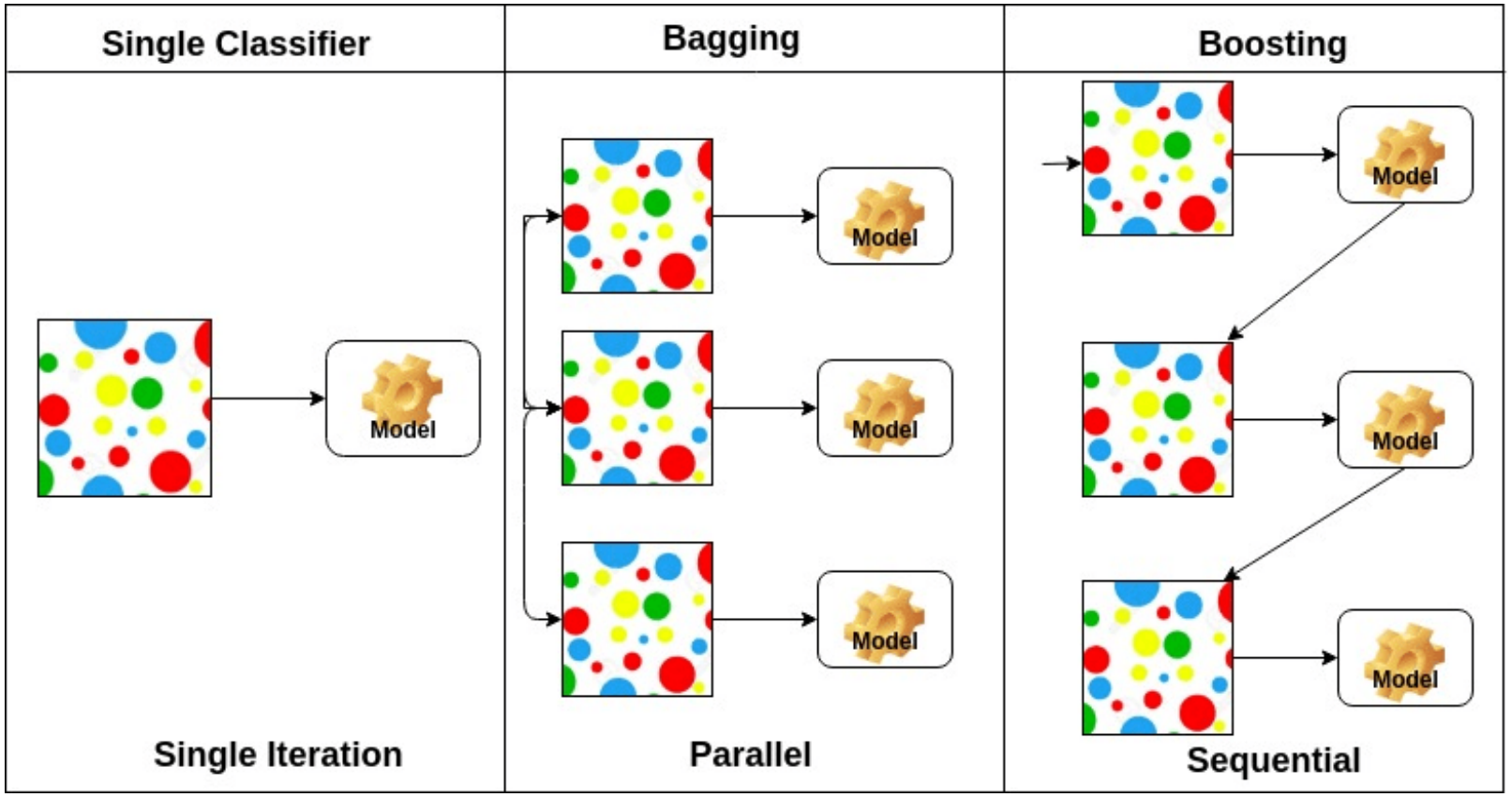
Disadvantages

- Issue of overfitting.
- **Low** predicted accuracy.
- Small change in the data can totally change the generated data.
- For the dominated classes, the decision tree learners build biased trees.

BOOSTING ALGORITHMS

- 1990 – Boost-by-majority algorithm (Freund)
- 1995 – AdaBoost (Freund & Schapire)
- 1997 – Generalized version of AdaBoost (Schapire & Singer)
- 2001 – AdaBoost in Face Detection (Viola & Jones)

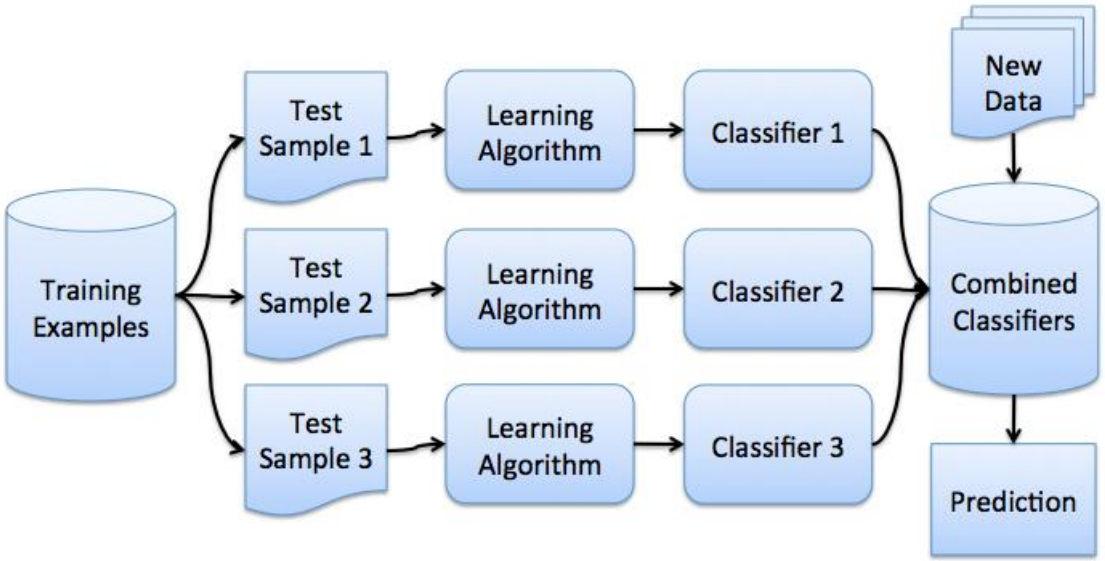
BOOSTING & BAGGING

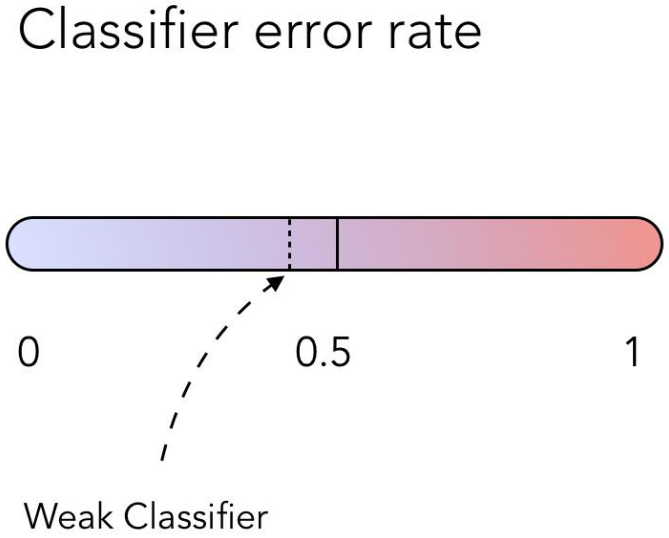


https://www.datacamp.com/community/tutorials/adaboost-classifier-python

BOOSTING ALGORITHMS

- Boosting trains a series a low performing algorithms, called weak learners, by adjusting the error metric over time.
- Weak learners are algorithms whose error rate is slightly under 50% as illustrated below.





https://prachimjoshi.wordpress.com/2015/07/23/bagging-and-boosting/

TYPES OF BOOSTING ALGORITHMS

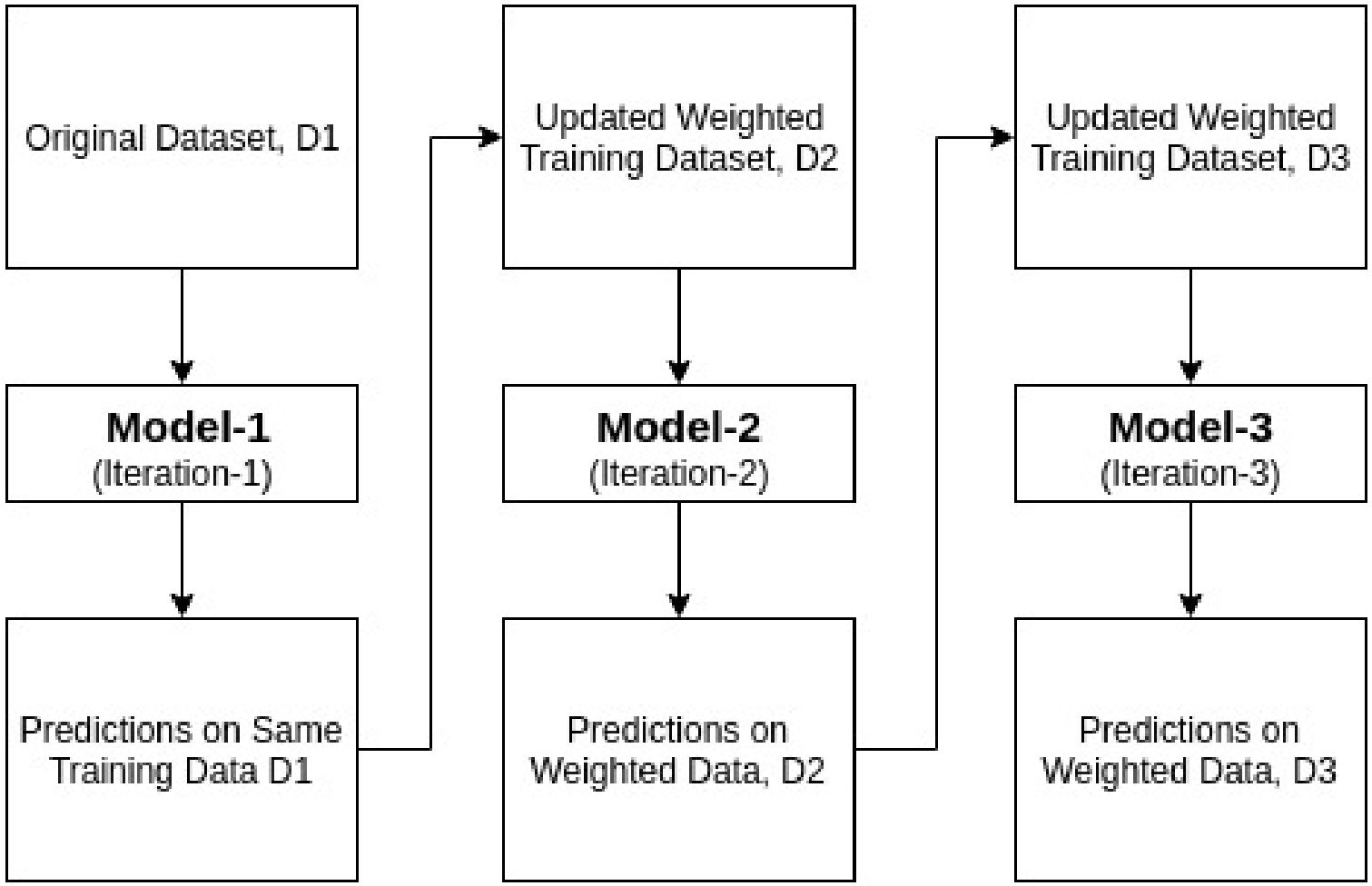
- AdaBoost (**Ada**ptive **Boost**ing)
- Gradient Tree Boosting
- XGBoost

- *GentleBoost*
- *LPBoost*
- *BrownBoost*
- CatBoost
- Lightgbm

ADABOOST

- AdaBoost = Adaptive Boosting (Schapire 1995)
- AdaBoost is an algorithm for constructing a ”strong” classifier as linear combination of “simple” “weak” classifiers.

$$H(x) = \sum_{t=1}^{T} \alpha_t h_t(x)$$


Original Dataset, D1
Updated Weighted Training Dataset, D2
Updated Weighted Training Dataset, D3
Model-1 (Iteration-1)
Model-2 (Iteration-2)
Model-3 (Iteration-3)
Predictions on Same Training Data D1
Predictions on Weighted Data, D2
Predictions on Weighted Data, D3

Algorithm 2 : AdaBoost.M1

1 **Input :** Dataset $D = \{(a_1, c_1), (a_2, c_2), \ldots, (a_N, c_N)\}$,
Base learner L and
Number of learning iteration T

2 Initialize equal weight to all training samples $w_i = 1/N$, $i = 1, 2, \ldots, N$

3 For $t = 1$ to T :

(a) Train a base learner h_t from D using D_t to training sample using w_i.
$h_t = L(D, D_t)$

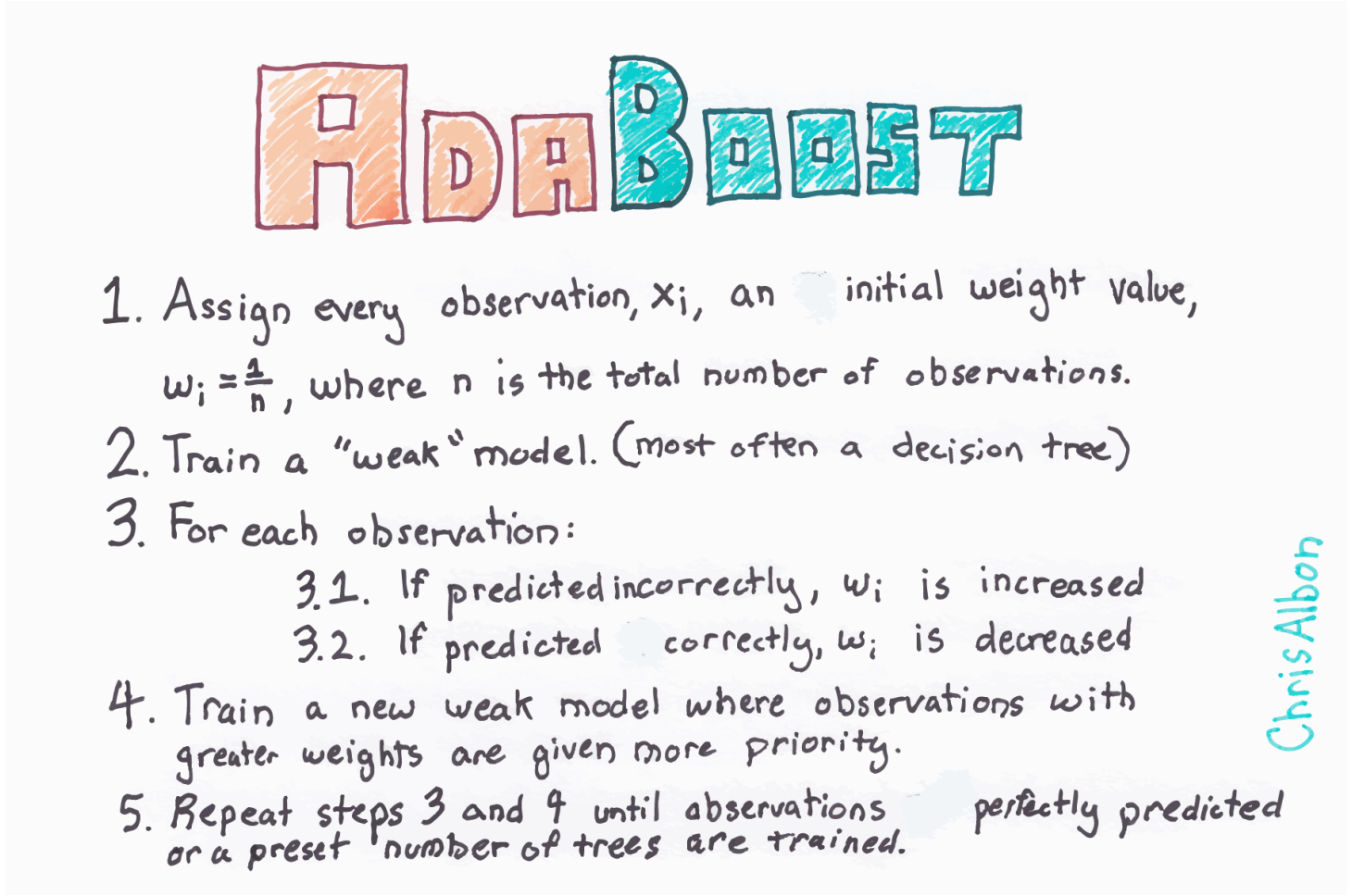
(b) Compute error of h_t as
$$err_t = \frac{\sum_{i=1}^{N} w_i I(h_t(a_i) \neq c_i)}{\sum_{i=1}^{N} w_i}$$

(c) Compute the weight of h_t as
$$\alpha_t = \log\left(\frac{1 - err_t}{err_t}\right)$$

(d) Set $w_i \leftarrow w_i . \exp[\alpha_t I(h_t(a_i) \neq c_i)]$

4 **Output :** $H(a) = \text{sign} \sum_{t=1}^{T} \alpha_t h_t(a)$

https://link.springer.com/chapter/10.1007/978-981-15-0633-8_22



https://chrisalbon.com/machine_learning/trees_and_forests/adaboost_classifier/

ADABOOST EXPLAINED & SIMPLE CODE

1. https://www.youtube.com/watch?v=LsK-xG1cLYA.
2. https://www.mygreatlearning.com/blog/adaboost-algorithm/
3. https://www.analyticsvidhya.com/blog/2021/09/adaboost-algorithm-a-complete-guide-for-beginners/

4. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.AdaBoostClassifier.html
5. https://www.datacamp.com/community/tutorials/adaboost-classifier-python

ADVANTAGES AND DISADVANTAGES OF ADABOOST

Advantages

- It is easier to use.
- It is not prone to overfitting.
- Can be used to improve the accuracy of your weak classifiers hence making it flexible.

Disadvantages

- Technique learns progressively, it is important to ensure that you have quality data.
- Extremely sensitive to Noisy data and outliers.

THANK YOU